No of Pages :4 **Course Code : 23MX11**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, DECEMBER 2024

MCA Semester : 1

23MX11 MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE

Time: 3 Hours **Maximum Marks:100**

INSTRUCTIONS:

1. Answer ALL questions. Each question carries 25 Marks.
2. Subdivision (a) carries 3 marks each, (b) carries 7 marks each and (c) carries 15 marks each.
3. Z, T and Chi-Square Table is permitted.
4. Graph sheet is to be given to the students.
5. Course Outcome : Table

Qn.1	CO1.	Qn.2	CO2.	Qn.3	CO3.	Qn.4	CO4.

COs	At the end of learning this course, students will be able to
CO1	Apply and solve problems based on set theory, relations, functions and logic. Solve the problems using Principle of Mathematical Induction.
CO2	Formulate and classify optimization problems and solve LPP problems using graphical, simplex, and two-phase methods. Understand the duality of LPP and solve special cases of LPP such as transportation problem and assignment problem.
CO3	Apply descriptive statistics, graphical presentations to data analysis and apply theory of probability, Bayes' rule to real world problems. Comprehend discrete and continuous random variables, joint random variables and apply them to solve real world problems
CO4	Apply linear regression and correlation to data analysis, undertake tests of hypotheses on single sample and two samples of data. Analyze time series data and forecast using moving average models.

Question **(BTL)**

1. a) A software company employs 100 computer programmers. Of them, 45 are proficient in Java, 30 in C#, 20 in Python, 6 in C# and Java, 1 in Java and Python, 5 in C# and Python, and just 1 in all three languages. How many are not proficient in any of these three programming languages? (L2)

b) i) Consider f : **N** → **N**, g : **N** → **N** and h : **N** → **N** defined as f (x) = 2x, g (y) = 3y + 4 and h (z) = sin z, ∀x, y and z in N. Show that h•(g•f) = (h•g)•f. (3) (L2)

ii) Consider f : {1, 2, 3} → {a, b, c} and g : {a, b, c} → {apple, ball, cat} defined as f(1) = a, f(2) = b, f (3) = c, g(a) = apple, g(b) = ball and g(c) = cat. Assume that f, g and g•f are invertible. Find f^{-1}, g^{-1} and $(g \bullet f)^{-1}$ and show that $(g \bullet f)^{-1} = f^{-1} \bullet g^{-1}$ (4) (L2)

c) i) Prove that the intersection of two equivalence relations R and S on a nonempty set is also an equivalence relation (5) (L3)

ii) There are n people in a room, where n is an integer greater than or equal to 1. Each person shakes hands once with every other person. Prove by mathematical induction that the number of handshakes is given by $\frac{n(n-1)}{2}$. (5) (L3)

iii) Show that the following proposition is a tautology : (5) (L3)

$[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$

2. a) Write the dual of the following Linear Programming problem. (L5)

Maximize $Z = 2x_1 + 3x_2 + x_3$

Subject to the following constraints

$4x_1 + 3x_2 + x_3 = 6$

$x_1 + 2x_2 + 5x_3 = 4$

$x_1, x_2, x_3 \geq 0$

b) A furniture dealer deals in only two items–tables and chairs. He has Rs 50,000 to invest and has storage space of at most 60 pieces. A table costs Rs 2500 and a chair Rs 500. He estimates that from the sale of one table, he can make a profit of Rs 250 and that from the sale of one chair a profit of Rs 75. He wants to know how many tables and chairs he should buy from the available money so as to maximize his total profit, assuming that he can sell all the items which he buys. Model the given problem mathematically and solve the same graphically. (L3)

c) i) a) Solve the following LPP using Simplex Method. (8) (L5)

Maximize $Z = 3x + 2y$

Subject to: $x \leq 4$; $2y \leq 12$; $3x + 2y \leq 18$ and $x, y \geq 0$

b) Consider the problem of assigning of five jobs to five persons. The assignment costs are given as follows. Determine the optimum assignment schedule. (7) (L4)

		JOBs				
		1	2	3	4	5
Persons	A	11	17	8	16	20
	B	9	7	12	6	15
	C	13	16	15	12	16
	D	21	24	17	28	26
	E	14	10	12	11	15

(OR)

ii) a) Solve the following LP problem using BIG M method. (10) (L5)

Minimize $Z = 4x_1 + x_2$

Subject to: $3x_1 + x_2 = 3$

$4x_1 + 3x_2 \geq 6$

$x_1 + 2x_2 \leq 4$ and $x_1, x_2 \geq 0$

b) Solve the following transportation problem using Least Cost method. (L4)

(5)

		Destinations			Supply
		I	II	III	
Sources	A	2	7	4	5
	B	3	3	1	8
	C	5	4	7	7
	D	1	6	2	14
Demand		7	9	18	

3 a) What is the Mean, Median and Mode of the following data set? (L2)

32, 6, 21, 10, 8, 11, 12, 36, 17, 16, 15, 18, 40, 24, 21, 23, 24, 24, 29, 16, 32, 31, 10, 30, 35, 32, 18, 39, 12, 20.

b) i) A box contains cards numbers between 0 and 100. A card is drawn from the box at random. What is the probability of getting a number on that card is a Fibonacci number. (3) (L3)

ii) Two dice are rolled. If it is known that at least one of the dice always shows 4, find the probability that the numbers appeared on the dice have sum 8. (4) (L3)

c) i) a) Three machines A1, A2, and A3 make 20%, 30%, and 50%, respectively, of the products. It is known that 1%, 4%, and 7% of the products made by each machine, respectively, are defective. If a finished product is randomly selected and if it is known that the selected product is defective, what is the probability that it is made by machine A1? (7) (L4)

b) Suppose that you have a fair 4-sided die, and let X be the random variable representing the value of the number rolled. Find the moment generating function for X. Use this moment generating function, compute the Mean and Variance of X. (8) (L5)

(OR)

ii) a) The following frequency distribution table shows the monthly consumption of electricity of 68 consumers of a locality. Find the median of the given data. (7) (L3)

65-85	85-105	105-125	125-145	145-165	165-185	185-205
4	5	13	20	14	8	4

b) Suppose that the diameter of a metal cylinder has the following Probability Density Function. (8)

$f(x) = 1.5 - 6(x - 50.0)^2$ for $49.5 \leq x \leq 50.5$

$= 0$, otherwise

1. Verify that, it is a valid Probability Density Function. (L5)

2. Calculate the probability that a metal cylinder has a diameter between 49.8 mm and 50.1 mm.

4 a) Given weekly demand with weights, what is the forecast for the 5th week? (L3)

Week	1	2	3	4	5
Demand	820	775	680	655	?
Weight	-----	0.1	0.2	0.7	

b) Find Pearson's coefficient of correlation between capital employed and profit obtained from the following data. (L5)

Capital Employed (Rs. In Crore)	10	20	30	40	50	60	70	80	90	100
Profit (Rs. In Crore)	2	4	8	5	10	15	14	20	22	50

c) 1) An online grocery shop claims that the mean delivery time is less than 120 minutes with a standard deviation of 30 minutes. A random sample of 49 orders is delivered with a mean of 100 minutes. At 95% confidence level, is there enough evidence to support the claim? (5) (L3)

2) A government agency reported that the population mean score on the quantitative portion of the Graduate Record Examination (GRE) General Test for students taking the exam between 2014 and 2017 was 558 ± 139 (μ±σ). (5) (L4)

Compute the sample test statistic at a 0.05 level of significance to take a decision on the hypothesis for the following scenarios.

i) Suppose, we select a sample of 100 students enrolled in an elite private school (n = 100) with a sample mean of 585 and hypothesize that students at this elite school will score higher than the general population.

ii) Suppose we select a sample of 100 students enrolled in a school with low funding and resources (n = 100) with a sample mean of 585 and hypothesize that students at this school will score lower than the general population. (L4)

3) In 120 throw of a single die, the following distribution of faces are obtained (5)

Face	1	2	3	4	5	6
Count	30	25	18	10	22	15

Do these results support the equal probability hypothesis at 0.05?

CO : COURSE OUTCOME; BTL : BLOOM'S TAXONOMY LEVEL

/END/

FD/TV